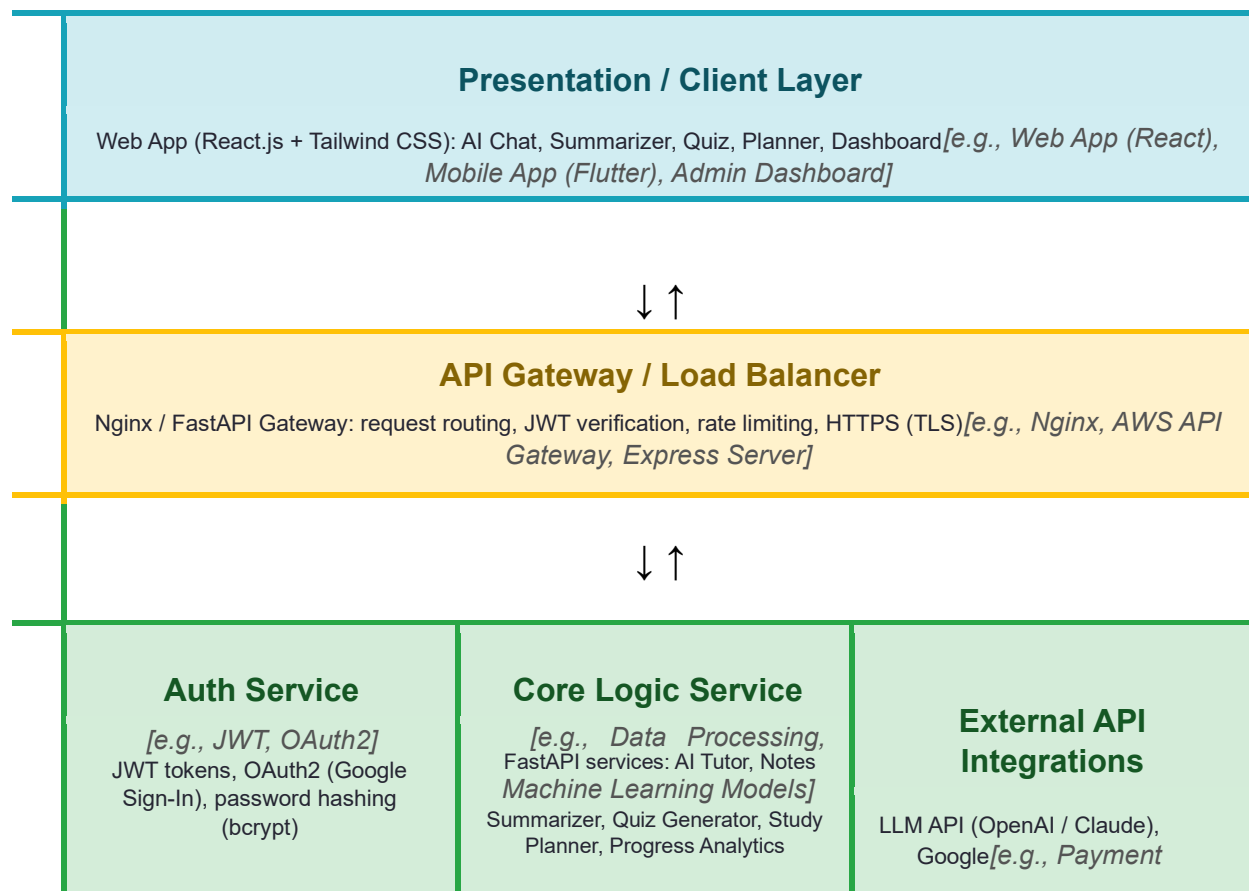


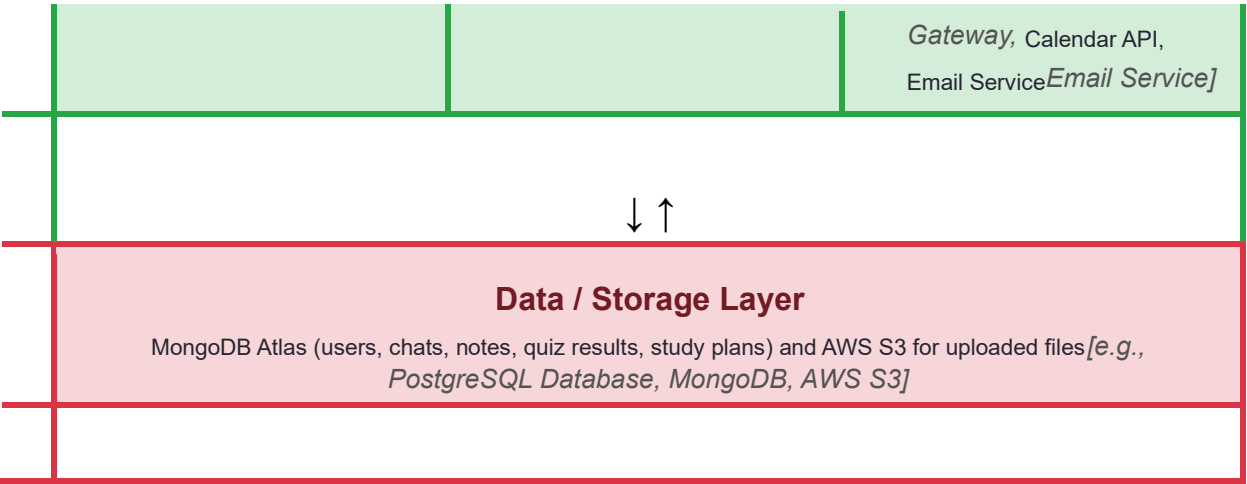
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-3887
Project Name	AI-STUDYBUDDY
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer[<i>Presentation Layer</i>]	Responsive UI for AI chat, notes upload, quizzes, planner [<i>Describe what this component does</i>] and progress dashboard.	React.js, Tailwind CSS[<i>List specific tech</i>]
API Gateway[<i>API Gateway</i>]	Single entry point that routes requests, verifies JWT tokens [<i>Describe what this component does</i>] and applies rate limits.	Nginx, FastAPI, HTTPS/TLS[<i>List specific tech</i>]
Microservices[<i>Microservices</i>]	Auth, AI Tutor, Summarizer, Quiz Generator, Planner and [<i>Describe what this component does</i>] Analytics services with the core logic.	Python, FastAPI, LLM API, [<i>List specific tech</i>] OAuth2/JWT
Database[<i>Database</i>]	Stores user profiles, chat history, notes, quiz results and [<i>Describe what this component does</i>] study plans; files in cloud storage.	MongoDB Atlas, AWS S3[<i>List specific tech</i>]